

4 (a) (i) 0.225 s and 0.525 s A1 [1]

(ii) period or $T = 0.30$ s and $\omega = 2\pi / T$ C1

$$\omega = 2\pi / 0.30$$

$$\omega = 21 \text{ rad s}^{-1}$$
A1 [2]

(iii) speed = ωx_0 or $\omega(x_0^2 - x^2)^{1/2}$ and $x = 0$ C1

$$= 20.9 \times 2.0 \times 10^{-2} = 0.42 \text{ m s}^{-1}$$
A1 [2]

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or

use of tangent method:

correct tangent shown on Fig. 4.2

(C1)

working e.g. $\Delta y / \Delta x$ leading to maximum speed in range $(0.38-0.46) \text{ m s}^{-1}$

(A1)

(b) sketch: reasonably shaped continuous oval/circle surrounding (0,0)

B1

curve passes through (0, 0.42) and (0, -0.42)

B1

curve passes through (2.0, 0) and (-2.0, 0)

B1

[3]